

Journal Club

Schulman, John, et al. "Proximal policy optimization algorithms." arXiv preprint arXiv:1707.06347 (2017).

Jongkook Choi

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Introduction

- ▶ Schulman, John, et al. "Proximal policy optimization algorithms." arXiv preprint arXiv:1707.06347 (2017).
- ▶ Cited by 38489
- ▶ John Schulman : Co-founder of OpenAI.

Policy Gradient

Conceptual Framework

- ▶ **Objective:** Optimal control of stochastic trajectories by finding a policy π_θ that maximizes the expected cumulative reward $J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} [\sum r_t]$.
- ▶ **From SDEs to RL:** Applying the Markov assumption to stochastic processes allows us to parameterize the control law as a stochastic policy $\pi_\theta(a_t|s_t)$.

The Policy Gradient Theorem The gradient of the objective is estimated using the log-derivative trick (likelihood ratio):

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{h=0}^{H-1} \nabla_\theta \ln \pi_\theta(a_h|s_h) R_h \right]$$

Supervised Learning

- ▶ **Data-Driven:** Track individual "particles" (data points) to sample sequences from a fixed dataset.
- ▶ **Objective:** Train the Transformer using Maximum Likelihood Estimation (MLE) until prediction accuracy is maximized.

Monte Carlo Approach (Policy Gradient/PPO)

- ▶ **Feynman-Kac:** The expected outcome of a stochastic process (SDE) can be estimated by sampling trajectories dictated by a "drift" term (PDE).
- ▶ **Objective:** Train the Transformer to optimize this drift (the policy), shifting the distribution toward high-reward trajectories. (so-called delta hedge)

Importance Sampling: Change of Measure

Motivation: Rare Events & High Variance

- ▶ Monte Carlo estimation suffers from high variance when target events are rare or the distribution has heavy tails.

Change of Measure To estimate $\mathbb{E}_P[f(X)]$ using samples from Q :

$$\mathbb{E}_P[f(X)] = \int f(x)p(x)dx = \int f(x)\frac{p(x)}{q(x)}q(x)dx = \mathbb{E}_Q\left[f(X)\frac{dP}{dQ}\right]$$

→ Efficient data usage is possible when P is close enough to the Q

Application in PPO

- ▶ **Target (P):** The new policy π_θ we want to optimize.
- ▶ **Proposal (Q):** The old policy $\pi_{\theta_{old}}$ used for data collection.
- ▶ **The Ratio (dP/dQ):** The importance weight

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}.$$

Example

- ▶ Calculate importance under Markovian assumption
$$\frac{dP}{dQ} = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}.$$
 (You can calculate it from the logits)
 - ▶ You could add transformer context as state, but there's high chance of you facing the curse of dimensionality.
- ▶ Sample trajectories using π_{old}
- ▶ Multiply weight using the importance,
- ▶ Estimate the expected gradient using the law of large numbers

Proximal policy optimization

"Proximal" : Keeping the new policy within a specific "proximity" or neighborhood of the old policy

PPO with Clipping and the Pessimistic Bound

For stability, PPO defines a pessimistic lower bound on the objective

$$L^{CLIP}(\theta) = \mathbb{E}_t [\min(r_t(\theta)(R_t - B_t), \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)(R_t - B_t))]$$

- ▶ Ignores improvement by extreme choice (gradient becomes zero when reward is greater than baseline and the importance is high)
- ▶ Ignores bad luck when the choice is normal (gradient becomes zero when reward is less than baseline and the importance is high)